

Table of contents

Detailed Course	1
Motivation and Context	2
What is Data Science?	2
Typical Data Pipeline	2
Representation Spaces	2
Formal Definition	2
Space Properties	3
Fundamental Questions	3
Curse of Dimensionality	4
1. Data Sparsity	4
2. Empty Sphere and Spiky Cube	5
3. Volume Distribution	7
4. The Gaussian Egg	8
5. Distance Concentration	10
6. Quasi-Orthogonality	11
7. Hubness	12
8. Concentration Inequalities	13
Blessing of Dimensionality	15
1. Laws of Large Numbers	15
2. Random Projections	16
Importance in Data Science	17
1. Sampling	17
2. Indexing	17
3. Learning	17
4. Why Does Data Science Still Work?	19
Global Summary	19
+ Positive Aspects	19
- Negative Aspects (Curse)	20
Solution	20

Detailed Course

Motivation and Context

What is Data Science?

Data Science studies:

- Modeling real-world phenomena as numerical (quantitative) data
- Geometric and statistical properties of this data
- Properties of the spaces in which this data resides
- Data analysis and the development of tools for this analysis
- Underlying assumptions in the design of these tools

Relation to Machine Learning: Data Science studies the representation spaces in which Machine Learning operates.

Quick Review: Motivation

Data Science = ML: Study of representation spaces where ML operates.

Pipeline: Process $\rightarrow$ Sensor $\rightarrow$ Raw Data $\rightarrow$ Representation (matrix in $\mathbb{R}^D$).

Focus on **geometric AND statistical** properties.

Typical Data Pipeline

General Flow: Process $\rightarrow$ Sensor $\rightarrow$ Raw Data $\rightarrow$ Representation

Concrete Examples:

- Camera $\rightarrow$ image pixels $\rightarrow$ matrix
 - Population $\rightarrow$ survey results $\rightarrow$ matrix
 - Text documents $\rightarrow$ word occurrences $\rightarrow$ matrix
 - Social network $\rightarrow$ relationships $\rightarrow$ matrix
 - Environment $\rightarrow$ measurements $\rightarrow$ matrix
-

Representation Spaces

Formal Definition

We obtain data $X = \{x_1, \dots, x_N\}$ where $x_i \in \Omega \subseteq \mathbb{R}^D$ for all $i \in \llbracket N \rrbracket$.

Statistical Perspective:

- X is a sample of size N from the underlying probability density function $f_X : \Omega \rightarrow \mathbb{R}^+$ with D variables
- We declare N i.i.d. random variables $\{X_i\}_{i \in [N]}$ where $X_i \sim f_X$
- Each data point x_i is a realization (sample) of X_i

Algebraic Perspective:

- We form the matrix $X = \begin{bmatrix} | & & | \\ x_1 & \cdots & x_N \\ | & & | \end{bmatrix} \in \mathbb{R}^{D \times N}$
- The columns $X_{\cdot j} = x_j$ are the data vectors

Space Properties

The representation space Ω is generally a subset of $\mathbb{R}^D$:

- It is a **vector space** that can be equipped with an inner product $\langle \cdot, \cdot \rangle$
- The inner product induces a **norm** $\| \cdot \|$
- The norm induces a **distance function** $d(\cdot, \cdot)$
- Favorable case: (Ω, d) is a **metric space** where learning can be performed

Note: Ω is also the basis of a measurable space on which random variables can be defined.

Quick Review: Representation Spaces

$X = \{x_1, \dots, x_N\}$ with $x_i \in \Omega \subseteq \mathbb{R}^D$.

Two perspectives:

- **Statistical:** Sample from f_X , i.i.d. variables
- **Algebraic:** Matrix $X \in \mathbb{R}^{D \times N}$

Structure: Vector space $\rightarrow$ inner product $\rightarrow$ norm $\rightarrow$ distance $\rightarrow$ metric space.

Fundamental Questions

What happens when:

- D increases? (dimensionality)
- N increases? (number of samples)

Impact on:

- Data modeling
- Machine learning

Curse of Dimensionality

1. Data Sparsity

Density Estimation Problem

Context:

- Population on hypercube $\Omega = [a, b]^D$
- Sample X from this population
- Objective: Estimate the empirical pdf via histogram

Method:

- Uniform quantization: b bins per dimension
- Desired quality: n samples per bin on average

Question: How many samples N are needed?

Answer:

- $D = 1 \rightarrow N \approx n \cdot b$
- $D = 2 \rightarrow N \approx n \cdot b^2$
- General $D \rightarrow N \approx n \cdot b^D$

Catastrophic Example: $b = 10, n = 10, D = 6 \rightarrow N \approx 10^7$ samples!

Estimation Quality with Fixed N

If we fix the number of samples at N :

Question: What estimation quality can we expect?

- $D = 1 \rightarrow n = \frac{N}{b} \rightarrow \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b}$
- General $D \rightarrow n = \frac{N}{b^D} \rightarrow \mathbb{E}[P(x_i \in \text{bin}_j)] \approx \frac{1}{b^D}$

Consequence: Most bins are likely empty

Quick Review: Data Sparsity

To estimate density with quality, need $N \approx n \cdot b^D$ samples (**exponential** growth).
If N is fixed: $n = N/b^D \rightarrow$ most bins **empty**.

Example: $D = 6$, $b = 10$, $n = 10 \rightarrow N \approx 10^7$.

2. Empty Sphere and Spiky Cube

Volume of the Hypersphere

Formula for a hypersphere $B_D(r)$ of radius r centered at the origin:

$$\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$$

Recurrence Relation (for $D > 1$):

$$\text{vol}(B_D(r)) = \frac{2\pi r^2}{D} \text{vol}(B_{D-2}(r))$$

Important Property:

$$\frac{\text{area}(B_D(r))}{\text{vol}(B_D(r))} = \frac{\pi^{1/2} D\Gamma(D/2)}{\Gamma(\frac{D+1}{2})}$$

Question: How does the volume of the sphere evolve as D increases?

Empty Sphere Phenomenon

Configuration: Hypersphere $B_D(r)$ inscribed in hypercube $C_D(r) = [-r, r]^D$.

Draw N uniform samples from the cube (i.e., from $\mathcal{U}(C_D(r))$).

Question: What proportion of samples falls within $B_D(r)$?

Calculation:

- $\text{vol}(B_D(r)) = \frac{2r^D \pi^{D/2}}{D\Gamma(D/2)}$
- $\text{vol}(C_D(r)) = (2r)^D$

Ratio:

$$\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1} \Gamma(D/2)}$$

Asymptotic Limit:

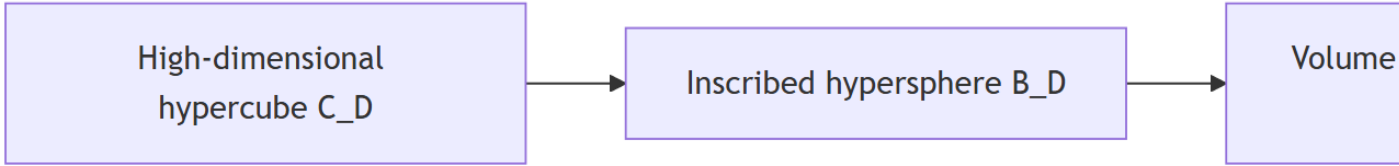
$$\lim_{D \rightarrow \infty} \frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = 0$$

Note: $\Gamma(n)$ behaves like $n! \sim n^n$

Consequences:

- All samples go to the part of the cube **outside the sphere** (the corners)
- The volume of the cube is **concentrated in its corners**
- All samples **escape from the center**
- The relative volume of the hypersphere tends to zero

The sphere is likely empty



Corners of the Hypercube

Configuration: Unit hypercube centered at $C_D(1) = [-\frac{1}{2}, \frac{1}{2}]^D$ with $\text{vol}(C_D(1)) = 1$.

Corner Properties:

- Number of corners: $n = 2^D$
- Each corner x_i has D coordinates: $|x_i(j)| = \frac{1}{2}$ for all j
- Distance to origin: $\|x_i\|_2 = \frac{\sqrt{D}}{2}$

Distance Table:

D	1	2	4	5	100
$\frac{\sqrt{D}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{5}}{2}$	5

Observation: The distance to corners **increases with** $\sqrt{D} \rightarrow$ corners move away from the center!

Quick Review: Empty Sphere & Spiky Cube

Sphere/cube ratio: $\frac{\text{vol}(B_D(r))}{\text{vol}(C_D(r))} = \frac{\pi^{D/2}}{D \cdot 2^{D-1} \Gamma(D/2)} \rightarrow 0$.

Volume concentrated at **corners** of the cube (center-corner distance = $\sqrt{D}/2$).

Uniform sampling $\rightarrow$ **empty** sphere.

3. Volume Distribution

For the Hypercube

Question: How is volume distributed in a shape of size r ?

Volume ratio between hypercubes of side r and $(1 - \varepsilon)r$ for $\varepsilon \in]0, 1[$:

$$\frac{\text{vol}(C_D((1 - \varepsilon)r))}{\text{vol}(C_D(r))} = \frac{((1 - \varepsilon)r)^D}{r^D} = (1 - \varepsilon)^D$$

Limit:

$$\lim_{D \rightarrow \infty} \frac{\text{vol}(C_D((1 - \varepsilon)r))}{\text{vol}(C_D(r))} = 0$$

For the Hypersphere

Volume ratio between hyperspheres of radius r and $(1 - \varepsilon)r$:

$$\frac{\text{vol}(B_D((1 - \varepsilon)r))}{\text{vol}(B_D(r))} = \frac{2((1 - \varepsilon)r)^D \pi^{D/2}}{D \Gamma(D/2)} \cdot \frac{D \Gamma(D/2)}{2r^D \pi^{D/2}} = (1 - \varepsilon)^D$$

Limit:

$$\lim_{D \rightarrow \infty} \frac{\text{vol}(B_D((1 - \varepsilon)r))}{\text{vol}(B_D(r))} = 0$$

Technical Note: $1 - \varepsilon \leq e^{-\varepsilon}$ so $(1 - \varepsilon)^D \leq e^{-D\varepsilon} \rightarrow 0$ as $D \rightarrow \infty$

Generalization to Any Volume

Principle: Decompose the volume into arbitrarily small (hyper-)cubic voxels.

Reduction: If we reduce the length of a voxel side by factor $(1 - \varepsilon)$:

- Voxel volume reduced by $(1 - \varepsilon)^D$
- Total volume reduced by $(1 - \varepsilon)^D$

Volume Ratio:

$$\lim_{D \rightarrow \infty} \frac{\text{vol}(V_D(r)) - \text{vol}(V_D((1 - \varepsilon)r))}{\text{vol}(V_D(r))} = \lim_{D \rightarrow \infty} 1 - (1 - \varepsilon)^D = 1$$

Fundamental Conclusion: Volume is concentrated near the surface of the shape

Quick Review: Volume Distribution

For any shape: Volume ratio $(1 - \varepsilon)^D \rightarrow 0$ as $D \rightarrow \infty$.

Method: Decomposition into hypercubic voxels.

Result: Volume is **concentrated at the surface** (difference $\rightarrow 1$).

4. The Gaussian Egg

Density Along the Radius

Question: Given a Normal distribution $\mathcal{N}(\mu, \Sigma)$, what is its density along the radius r ?

Context: The coordinates are D i.i.d. random variables $X_i \sim \mathcal{N}(0, 1)$.

Integration along r = take the norm. R is the random variable associated with the radius:

$$R^2 = \|X\|_2^2 = \sum_{i=1}^D X_i^2 \sim \chi^2(D)$$

Expectation:

$$\mathbb{E}[R^2] = D \implies \mathbb{E}[R] \approx \sqrt{D}$$

We can consider the centered and scaled distribution (i.e., $\mathcal{N}(0, \text{Id}_D)$) with $\sigma = 1$.

Density is concentrated at radius $\sigma\sqrt{D}$



Gaussian Annulus Theorem

Given $X \sim \mathcal{N}(0, \text{Id}_D)$, for any $\beta \leq \sqrt{D}$:

$$P\left[\left|\|X\|_2 - \sqrt{D}\right| > \beta\right] \leq 3e^{-c\beta^2}$$

where $c > 0$ is a constant.

Interpretation: The norm is **exponentially concentrated** around $\sqrt{D}$.

Alternative Formulation:

$$P\left[\left|\frac{\|X\|}{\sqrt{D}} - 1\right| \leq \varepsilon\right] \approx 1$$

Almost all density is in a **thin ring** around radius $\sqrt{D}$

Quick Review: Gaussian Egg

For $X \sim \mathcal{N}(0, \text{Id}_D)$: $R^2 = \|X\|_2^2 \sim \chi^2(D)$ so $\mathbb{E}[R^2] = D$.

Gaussian Annulus Theorem: $P[|\|X\|_2 - \sqrt{D}| > \beta] \leq 3e^{-c\beta^2}$.

Density concentrated at radius $\sigma\sqrt{D}$ (**thin ring**).

5. Distance Concentration

Context: Data $X \subset \Omega$, select a data point x and its k nearest neighbors $V_X^k(x)$.

Let $D_{\min}$ and $D_{\max}$ be the random variables associated with the distances to the nearest and farthest neighbors among the k neighbors.

Concentration Theorem

Theorem: If $\lim_{D \rightarrow \infty} \text{Var}\left[\frac{d(x,y)^p}{\mathbb{E}[d(x,y)^p]}\right] = 0$ for $0 < p < \infty$, then for any $\varepsilon > 0$:

$$\lim_{D \rightarrow \infty} P\left[\frac{D_{\max} - D_{\min}}{D_{\min}} \leq \varepsilon\right] = 1$$

Consequences:

- All neighbors appear at distance $D_{\min}$ from x
- Discriminative power **decreases exponentially**
- KNN (*k-nearest neighbors*) and ε NN structures become **non-significant**

Quick Review: Distance Concentration

If normalized distance variance $\rightarrow 0$, then $\frac{D_{\max} - D_{\min}}{D_{\min}} \rightarrow 0$.

All neighbors are **equidistant**.

KNN and ε NN **lose their meaning**.

6. Quasi-Orthogonality

Concentration of Angles Between Random Vectors

Question: When drawing vectors from a D -dimensional space, what is their expected angle?

Method: Declare 4 i.i.d. random variables W, X, Y, Z on $\Omega \subset \mathbb{R}^D$ and compute:

$$\mathbb{E}[\cos(\angle(X - Y, Z - W))] = \mathbb{E}\left[\frac{\langle X - Y, Z - W \rangle}{\|X - Y\| \cdot \|Z - W\|}\right]$$

Since W, X, Y, Z are i.i.d., $X - Y$ and $Z - W$ are also i.i.d. and centered:

$$\mathbb{E}(X - Y) = \mathbb{E}(Z - W) = 0$$

The numerator is $\text{cov}(X - Y, Z - W) = 0$, the denominator is a normalizer.

Result: $\mathbb{E}[\cos(\theta)] = 0 \implies \theta \sim \frac{\pi}{2}$

Alternative:

$$\langle X - Y, Z - W \rangle = \langle X, Z \rangle - \langle X, W \rangle - \langle Y, Z \rangle + \langle Y, W \rangle$$

In expectation, all terms cancel because W, X, Y, Z are i.i.d.

Conclusion:

- Every random triangle is a **right triangle**
- Two random vectors are **quasi-orthogonal**

Concentration of Angles in a Triangle

Question: What is the expected angle in a random triangle?

Method: Declare 3 i.i.d. random variables X, Y, Z and compute for an angle:

$$\mathbb{E}[\cos(\angle(X - Y, Z - Y))] = \mathbb{E}\left[\frac{\langle X - Y, Z - Y \rangle}{\|X - Y\| \cdot \|Z - Y\|}\right]$$

Note: $X - Y$ and $Z - Y$ are **no longer independent!**

Calculation (simplified):

$$\frac{\langle X - Y, Z - Y \rangle}{\langle X - Y, X - Y \rangle} = \frac{\langle X, Z \rangle - \langle X, Y \rangle - \langle Y, Z \rangle + \langle Y, Y \rangle}{\langle X, X \rangle + \langle Y, Y \rangle - 2\langle X, Y \rangle}$$

In expectation:

$$= \frac{\langle Y, Y \rangle - \langle X, Y \rangle}{2\langle Y, Y \rangle - 2\langle X, Y \rangle} = \frac{1}{2}$$

Result: $\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \sim \frac{\pi}{3}$

Conclusion: Every random triangle is an **equilateral triangle**

Impact for the Sphere

Given the unit ball B_D , if X is a random point on its surface and Y another point, let $\theta = \angle(X, Y)$:

- If Y sampled from Ω : $\mathbb{E}[\cos(\theta)] = \frac{1}{2} \implies \theta \sim \frac{\pi}{3}$
- If Y sampled from B_D : $\mathbb{E}[\cos(\theta)] = 0 \implies \theta \sim \frac{\pi}{2}$

The volume of the sphere is contained at its **equator** (for any north direction X)

→ **Alternative proof** for volume concentration at the sphere's surface

Note: To sample on the unit hypersphere, sample from a centrally symmetric distribution (e.g., Gaussian) and normalize.

Quick Review: Quasi-Orthogonality

2 random vectors: $\mathbb{E}[\cos(\theta)] = 0 \rightarrow \theta \sim \pi/2$ (quasi-orthogonal).

Random triangle: $\mathbb{E}[\cos(\theta)] = 1/2 \rightarrow \theta \sim \pi/3$ (equilateral).

Sphere: Volume at the equator.

7. Hubness

Definition

Question: How many times does a sample appear as a k -nearest neighbor of another data point?

Given X , define:

$$r_{ik}(x_j) = \begin{cases} 1 & \text{if } x_j \in V_X^k(x_i) \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$n_k(x_j) = \sum_i r_{ik}(x_j)$$

Result: The distribution of n_k values is **left-skewed**.

Consequence: A small number of samples appear in the neighborhood of **many** other samples.

These **hubs** are close to all other samples

Empirical Visualization: For 20-NN with $D = 100$ and 1000 uniform samples on 50 bins, a strong asymmetry in the distribution of n_k is observed.

Quick Review: Hubness

$n_k(x_j)$ = number of times x_j is a k-NN of another point.

Asymmetric distribution: few **hubs** appear in many neighborhoods.

These hubs are “close” to all.

8. Concentration Inequalities

Fundamental Inequalities

Markov's Inequality: For a non-negative random variable X :

$$P(X > \varepsilon) \leq \frac{\mathbb{E}[X]}{\varepsilon}$$

Chebyshev's Inequality: For a random variable with mean μ and finite variance σ^2 :

$$P(|X - \mu| > \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}$$

Hoeffding's Inequality

Statement: If $\{X_i\}_{i \in [D]}$ are independent random variables with $P(a \leq X_i \leq b) = 1$ and $\mathbb{E}[X_i] = \mu$, then for any $\varepsilon > 0$:

$$P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$$

where $\bar{X} = \frac{1}{D} \sum_{i=1}^D X_i$

Note: Example of PAC (*Probably Approximately Correct*) analysis: probability that an event occurs approximately.

Explanation of Distance Concentration

Hoeffding's inequality explains:

- When aggregating independent variables, the aggregate **converges to their common expectation**
- The structure $\sum_{i=1}^D (\cdot)$ corresponds to the calculation of Minkowski distances:

$$d_p(x, y) = \left(\sum_{i=1}^D |x[i] - y[i]|^p \right)^{1/p}$$

- Hoeffding's inequality formally explains why distance values **concentrate**
- Although counterintuitive that all neighbors are equidistant, it is intuitive that for large D , extreme values (small or large) of $d_p(x, y)$ become unlikely: for $d_p(x, y)$ to be small (large), **all** D coordinates $x[i]$ and $y[i]$ must be (dis)similar. As D increases, this becomes unlikely.

Quick Review: Concentration Inequalities

Hoeffding: $P(|\bar{X} - \mu| > \varepsilon) \leq 2e^{-2D\varepsilon^2/(b-a)^2}$.

Aggregation of independent variables $\rightarrow$ convergence to μ .

Structure $\sum_{i=1}^D (\cdot)$ in distances $\rightarrow$ **concentration of distances**.

Blessing of Dimensionality

1. Laws of Large Numbers

Weak Law of Large Numbers (WLLN)

Statement: Given a set of D i.i.d. random variables $\{X_i\}_{i \in [D]}$ with $\mathbb{E}[X_i] = \mu$, then for any $\varepsilon > 0$:

$$\lim_{D \rightarrow \infty} P(|\bar{X} - \mu| < \varepsilon) = 1$$

where $\bar{X} = \frac{1}{D} \sum_{i=1}^D X_i$

(Can be proven from Chebyshev's inequality)

Interpretation:

- The WLLN states that the **empirical mean** $\bar{X}$ converges (in probability) to the **true expectation** μ as the number of samples increases
- Fundamentally consistent with the frequentist approach to probabilities
- It is the **fundamental basis** of estimation theory (and density estimation)

Central Limit Theorem (CLT)

The form and rate of convergence are given by the CLT:

$$Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$$

if $\text{Var}(X_i) = \sigma^2$

Quick Review: Laws of Large Numbers

WLLN: $\bar{X} \xrightarrow{P} \mu$ as $D \rightarrow \infty$.

Basis for density estimation.

CLT: $Z_D = \frac{\sqrt{D}}{\sigma}(\bar{X} - \mu) \sim \mathcal{N}(0, 1)$ (convergence rate).

2. Random Projections

Principle

Recall: Two random vectors u and v are likely orthogonal.

Quasi-orthogonality: $\langle u, v \rangle = \varepsilon$ (small).

Key Property: **Exponentially many** quasi-orthogonal directions in high-dimensional spaces.

Application: Create random bases (quasi-orthonormal) for signal decomposition and coding.

Johnson-Lindenstrauss Lemma

Statement: Given $X \subset \Omega$, there exists a linear map $p : \mathbb{R}^D \rightarrow \mathbb{R}^d$ such that, with $0 < \varepsilon < 1$ and for all $i, j \in \llbracket N \rrbracket$:

$$(1 - \varepsilon)\|x_i - x_j\| \leq \|p(x_i) - p(x_j)\| \leq (1 + \varepsilon)\|x_i - x_j\|$$

and

$$d = O\left(\frac{\log N}{\varepsilon^2}\right) \leftarrow \text{Does NOT depend on } D!$$

Applications:

- *Locally Sensitive Hashing* (LSH) for approximate nearest neighbor (ANN) search
- **Caveat:** Convergence rate slower than most concentrations

Quick Review: Random Projections

Principle: Exponentially many quasi-orthogonal directions in high dimensions.

Johnson-Lindenstrauss: Projection $\mathbb{R}^D \rightarrow \mathbb{R}^d$ preserving distances (up to ε) with $d = O(\log N / \varepsilon^2)$ (independent of D !).

Application: LSH for ANN.

Importance in Data Science

1. Sampling

Consequences of Sparsity:

- An **exponential number of samples** is needed to cover a high-dimensional space
- Concentration inequalities (including WLLN and CLT) are **precision indicators** for density estimation
- In practical processes like rejection sampling, the **rejection rate** becomes exponentially close to 100% (cf. empty sphere)

2. Indexing

Exploiting Structure:

- Indexing exploits data structure to anticipate which parts of space Ω to visit at query time
- If all neighbors appear at **equal distance** from the center, the structure **disappears**

Spatial Partitioning Techniques (e.g., *kd-trees*):

- Attempt to partition space so that each data point is uniquely identified within a cell intersection (paths in the search tree)
- With distance concentration, cell size reaches the **sphere's radius** to cover the entire space

Search becomes **exhaustive** complexity $O(N)$

- *Hubness* makes some data points answers to **almost all queries**

3. Learning

Function Approximation

Machine Learning = discovering **function approximators**.

Problem: Given $\phi : X \rightarrow Y$ the true function mapping data X to labels Y , ML seeks an approximation $\phi_\theta \in \mathcal{F}$ of ϕ minimizing error (risk) on samples (train/test).

Theoretical Bound

If $\mathcal{F}$ is a class of c -uniformly Lipschitz functions on Ω :

$$\sup_{\phi_\theta \in \mathcal{F}} \|\phi_\theta - \phi\|_\infty \geq \frac{c\sqrt{D}N^{-1/D}}{2} \sqrt{\frac{2}{\pi e}} \left(1 + O\left(\frac{\log D}{D}\right)\right)$$

If we allow an error $c\varepsilon$ for $\varepsilon > 0$, the required number of training examples N is:

$$N \geq \frac{\varepsilon^{-D} D^{D/2}}{(2\pi e)^{D/2}}$$

Concrete Examples:

- $c = 1, \varepsilon = 0.1, D = 3 \rightarrow N \geq 74$
- $c = 1, \varepsilon = 0.1, D = 10 \rightarrow N \geq 687 \times 10^6$
- $c = 1, \varepsilon = 0.1, D = 15 \rightarrow N \geq 377 \times 10^{12}$
- $c = 1, \varepsilon = 0.01, D = 10 \rightarrow N \geq 6.9 \times 10^{18}$

Realistic Case in ML:

- ML data: typically hundreds of features ($D \sim O(10^2)$)
- Typical desired error: $\varepsilon \approx 10^{-4}$

$$N \geq \frac{10^{4 \cdot 100} \cdot 100^{50}}{17.07^{50}} \approx 10^{400}$$

Conclusion: This is **impossible** in practice!

Quick Review: Importance in Data Science

Sampling: Exponential sample requirement, rejection rate $\rightarrow 100\%$.

Indexing: Structure disappears, kd-trees $\rightarrow$ complexity $O(N)$, hubness.

Learning: $N \geq \varepsilon^{-D} D^{D/2} / (2\pi e)^{D/2} \rightarrow$ impossible for large D and small ε .

4. Why Does Data Science Still Work?

Question: Why do data science processes work despite these theoretical barriers?

Assumptions in the Above Analysis:

- Most analyses assume a **uniform** (or isotropic – Normal, ...) distribution of data
- Most analyses assume **independent** features

Reality:

- **Real data does NOT populate the feature space uniformly** (if features are informative)
- **Features show correlations**

In practice, data populates **subspaces** of Ω with **lower local intrinsic dimensionality**

Solutions:

- **Decorrelate features** (e.g., via linear latent space)
- **Discover data subspaces** (clustering, nonlinear dimensionality reduction)

Quick Review: Why It Works

Analyses assume **uniform** distribution and **independent** features.

Reality: Data lies on **subspaces** of lower intrinsic dimensionality.

Solutions: Decorrelation, clustering, dimensionality reduction.

Global Summary

High-dimensional geometry ($D \gtrsim 10$) is **different** from familiar 3D geometry.

+ Positive Aspects

- **Concentration inequalities** are the basis of statistical estimation
- **Laws of Large Numbers** enable density estimation
- **Random projections** exploit quasi-orthogonality

- Negative Aspects (Curse)

- Data becomes very **sparse** unless using exponentially more data
- Even with dense data, **distance and angle values concentrate**
 - From a query, all neighbors are **equidistant**
 - From a direction, everything else appears **orthogonal**
- Density concentrates on **narrow radii** $\rightarrow$ biased sampling
- **Hubs** are part of all class/query responses

Solution

Since this is by construction, the only solution is to operate in **low-dimensional spaces** in some way:

- Dimensionality reduction
- Subspace discovery
- Feature decorrelation

Quick Review: Summary

High dimension ($D \gtrsim 10$) 3D geometry.

: Concentration inequalities for estimation, random projections.

: Data sparsity, distance/angle concentration, hubs.

Solution: Operate in **low dimension**.